

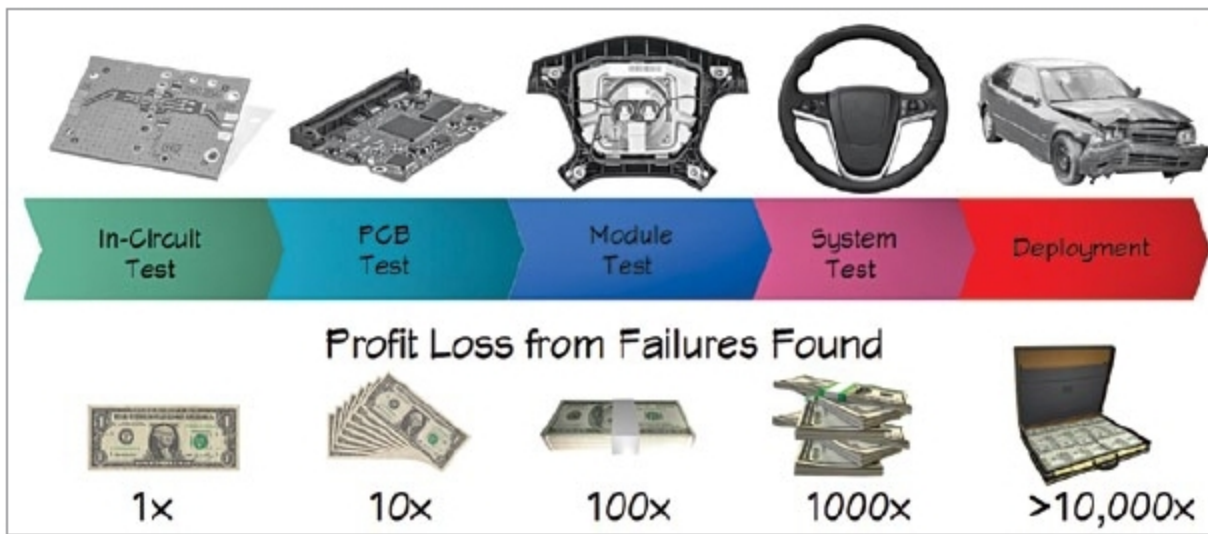


Fig. 2: The 10x rule



Fig. 3: Lean Six Sigma approach

certain rules where violation justifies investigation, depending on how far the observations are from ranges of standard deviations. One problematic feature of WECO is that it, on average, triggers a false alarm every 9175th unrelated measurement.

False alarms everywhere!

Let us say, you have an annual production output of 10,000 units. Each gets tested through five different processes. Each process has an average of 25 measurements. Combining these you get up to 62 false alarms per day on average, assuming 220 working days per year.

Let us repeat that. Assuming you, against all odds and reason, can remove common cause variations. You would still receive 62 alarms every day. People receiving 62 emails per day from a single source would likely block these, leaving potentially important announcements unacknowledged, with no follow up.

SPC-savvy users will likely argue that there are ways to reduce this by

new and improved analytical methods. They might say, "There are Nelson Rules, we have AIAG, you should definitely use Juran Rules. You need to identify auto-correlation structures to reduce the number of false alarms. What about the ground-breaking state-of-the-art chart developed in early 2000s? Given it a go yet?"

Even if you manage to reduce the number of false alarms to five per day, could it represent a strategic alarming system? Adding actual process dynamics to the mix, can SPC produce a system that manufacturing managers rely on? One that keeps their concerns and ulcers at bay?

Enter KPIs

What most people do is make assumptions based on a limited set of important parameters to monitor, and carefully track these by plotting these on their control charts, X-mR charts or whatever they use to try and separate the wheat from the chaff. These KPIs are often captured and analysed downstream in the manufacturing process, after multiple units are combined into a system.

An obvious consequence of this is that problems are not detected where these happen, as these happen. The origin could easily come from one of the components or processes upstream, manufactured one month ago in a batch that by now has reached 50,000 units.

A cost-failure relationship known as the 10x rule says that, for each step in the manufacturing process a failure is allowed to continue, the cost of fixing it increases by a factor of 10. A failure found at system level can mean

that technicians will need to rip apart the product, an act that gives opportunities for new defects.

If failure is allowed to reach the field, cost implications can be catastrophic. There are multiple examples from modern times where firms had to declare bankruptcy or protection against it due to the prospect of massive recalls. A recent example is Takata filing for bankruptcy after a massive recall of airbag components, which may exceed 100 million units.

One of the big inherent flaws of SPC, according to standards of modern approaches such as Lean Six Sigma, is that it makes assumptions of where problems are coming from. This is an obvious consequence of assuming stability in what are highly dynamic factors. Trending and tracking a limited set of KPIs only enhance this flaw. This again kicks off improvement initiatives that are likely to fail at focusing on the most pressing or cost-efficient issues.

A modern approach

All this is accounted for in modern methods for quality management. In electronics manufacturing, this starts with an honest recognition and monitoring of first pass yield (FPY)—true FPY to be precise. By true it means that any kind of failure must be accounted for, even if it only came from something as simple as the test operator forgetting to plug in a cable.

Every test after the first represents waste, resources the company could have spent better elsewhere. True FPY represents the single most important KPI, still most OEMs have no real clue what their's is.

Knowing your FPY, you can break this down in parallel across different products, product families, factories, stations, fixtures, operators and test operation. Having this data available in real time as dashboards gives you a powerful captain's view. It lets you quickly drill down to understand what the real origin of poor performance is and make interventions based on economic reasoning.

Allocating this insight as live dashboards to all involved stakeholders also contributes to enhanced accountability of quality. A good